

MENVEO

Meningococcal Group A, C, W135 and Y Conjugate Vaccine

1. NAME OF THE MEDICINAL PRODUCT

Menveo powder and solution for solution for injection
Meningococcal Group A, C, W135 and Y conjugate vaccine

2. QUALITATIVE AND QUANTITATIVE COMPOSITION

One dose (0.5 ml of the reconstituted vaccine) contains:

(Originally contained in the powder)

- | | |
|---|-------------------------|
| • Meningococcal group A oligosaccharide | 10 micrograms |
| Conjugated to <i>Corynebacterium diphtheriae</i> CRM ₁₉₇ protein | 16.7 to 33.3 micrograms |

(Originally contained in the solution)

- | | |
|---|------------------------|
| • Meningococcal group C oligosaccharide | 5 micrograms |
| Conjugated to <i>Corynebacterium diphtheriae</i> CRM ₁₉₇ protein | 7.1 to 12.5 micrograms |
| • Meningococcal group W135 oligosaccharide | 5 micrograms |
| Conjugated to <i>Corynebacterium diphtheriae</i> CRM ₁₉₇ protein | 3.3 to 8.3 micrograms |
| • Meningococcal group Y oligosaccharide | 5 micrograms |
| Conjugated to <i>Corynebacterium diphtheriae</i> CRM ₁₉₇ protein | 5.6 to 10.0 micrograms |

For a full list of excipients, see section 6.1.

3. PHARMACEUTICAL FORM

Powder and solution for solution for injection (powder and solution for injection).

The powder is a white to off- white cake.

The solution is a colourless clear solution.

4. CLINICAL PARTICULARS

4.1 Therapeutic indications

Menveo is indicated for active immunization of children (from 2 years of age) and adults at risk of exposure to *Neisseria meningitidis* groups A, C, W135 and Y, to prevent invasive disease. The use of this vaccine should be in accordance with official recommendations.

4.2 Posology and method of administration

Posology

Adults and children (above 5 years of age)

Menveo should be administered as a single 0.5 ml injection.

Children (2-5 years of age)

Menveo should be administered as a single 0.5 ml injection. A second dose may be administered 2 months after the first dose.

Elderly

There are limited data in individuals aged 56-65 and there are no data in individuals aged >65 years.

Booster vaccination

Long-term antibody persistence data following vaccination with Menveo are available up to 5 years after vaccination (see section 4.4 and 5.1).

Menveo may be given as a booster dose in subjects who have previously received primary vaccination with Menveo, other conjugated meningococcal vaccine or meningococcal unconjugated polysaccharide vaccine. The need for and timing of a booster dose in subjects previously vaccinated with Menveo is to be defined based on national recommendations.

Paediatric population (under 2 years of age)

The safety and efficacy of Menveo in children under 2 years of age has not yet been established. Currently available data are described in section 5.1 but no recommendation on a posology can be made.

Method of administration

Menveo is given as an intramuscular injection, preferably into the deltoid muscle. It must not be administered intravascularly, subcutaneously or intradermally.

Separate injection sites must be used if more than one vaccine is being administered at the same time.

For instructions on preparation and reconstitution of the product, see section 6.6.

4.3 Contraindications

Hypersensitivity to the active substance or to any of the excipients listed in section 6.1, or diphtheria toxoid (CRM₁₉₇), or a life-threatening reaction after previous administration of a vaccine containing similar components (see section 4.4).

As with other vaccines, Menveo should be postponed in individuals suffering from an acute severe febrile illness. The presence of a minor infection is not a contraindication.

4.4 Special warnings and precautions for use

Before the injection of any vaccine, the person responsible for administration must take all precautions known for the prevention of allergic or any other reactions including thorough medical history and current health status.

As with all injectable vaccines, appropriate medical treatment and supervision must always be readily available in case of a rare anaphylactic event following administration of the vaccine. Anxiety-related reactions, including vasovagal reactions (syncope), hyperventilation or stress-

related reactions may occur in association with vaccination as a psychogenic response to the needle injection (see Adverse Reactions). It is important that procedures are in place to avoid injury from fainting.

Menveo should under no circumstances be administered intravascularly.

Menveo will not protect against infections caused by any other serogroups of *N. meningitidis* not present in the vaccine.

As with any vaccine, a protective immune response may not be elicited in all vaccinees (see section 5.1).

Studies with Menveo have shown a waning of serum bactericidal antibody titers against serogroup A when using human complement in the assay (hSBA) (see section 5.1). The clinical relevance of the waning of hSBA serogroup A antibody titers is unknown. If an individual is expected to be at particular risk of exposure to Men A and received a dose of Menveo more than approximately one year previously, consideration may be given to administering a booster dose.

There are no data on the applicability of the vaccine for post-exposure prophylaxis.

In immunocompromised individuals, vaccination may not result in an appropriate protective antibody response. While Human Immunodeficiency Virus (HIV) infection is not a contraindication, Menveo has not been specifically evaluated in immunocompromised people. Individuals with complement deficiencies and individuals with functional or anatomical asplenia may not mount an immune response to meningococcal group A, C, W135 and Y conjugate vaccines.

Individuals with familial complement deficiencies (for example, C3 or C5 deficiencies) and individuals receiving treatments that inhibit terminal complement activation (for example, eculizumab) are at increased risk for invasive disease caused by *Neisseria meningitidis* group A, C, W-135 and Y, even if they develop antibodies following vaccination with Menveo.

Menveo has not been evaluated in persons with thrombocytopenia, bleeding disorders or that are receiving anticoagulant therapy, because of the risk of haematoma. The risk- benefit ratio for persons at risk of haematoma following intramuscular injection must be evaluated by health care professionals.

4.5 Interaction with other medicinal products and other forms of interaction

Menveo can be given concomitantly with any of the following vaccines: monovalent and combined hepatitis A and B, yellow fever, typhoid fever (Vi polysaccharide), Japanese encephalitis, rabies and meningococcal group B (Bexsero).

In adolescents (11 to 18 years of age), Menveo has been evaluated in two co-administration studies with either Tetanus, Reduced Diphtheria and Acellular Pertussis Vaccine, Adsorbed (Tdap) alone or Tdap and Human Papillomavirus Quadrivalent (Types 6, 11, 16 and 18) Vaccine, Recombinant (HPV), both of which support the co-administration of the vaccines.

There was no evidence of an increased rate of reactogenicity or change in the safety profile

of the vaccines in either study. Antibody responses to Menveo and the diphtheria, tetanus or HPV vaccine components were not negatively affected by co-administration.

The administration of Menveo one month after Tdap resulted in statistically significantly lower group W135 seroresponses. Since there was no direct impact on the seroprotection rate, the clinical consequences are presently unknown.

There was evidence of some suppression of antibody response to two of the three pertussis antigens. The clinical relevance of this observation is unknown. After vaccination, over 97% of subjects had detectable pertussis titers to all three pertussis antigens.

For children 2 to 10 years of age, no data are available for evaluating safety and immunogenicity of other childhood vaccines when administered concomitantly with Menveo.

Concomitant administration of Menveo and other vaccines than those listed above has not been studied. Concomitant vaccines should always be administered at separate injection sites and preferably contralateral. It should be checked if the adverse reactions may be intensified by any co-administration.

If a vaccine recipient is undergoing immunosuppressant treatment, the immunological response may be diminished.

4.6 Fertility, pregnancy and lactation

Insufficient clinical data on exposed pregnancies are available.

In non-clinical studies, Menveo had no direct or indirect harmful effects with respect to pregnancy, embryonal/foetal development, parturition or postnatal development. Considering the severity of invasive meningococcal disease caused by *Neisseria meningitidis* serogroups A, C, W and Y, pregnancy should not preclude vaccination when the risk of exposure is clearly defined.

Although insufficient clinical data on the use of Menveo during breast-feeding are available, it is unlikely that secreted antibodies in milk would be harmful when ingested by a breastfed infant. Therefore, Menveo may be used during breast feeding.

4.7 Effects on ability to drive and use machines

No studies on the effects on the ability to drive and use machines have been performed. Dizziness has been very rarely reported following vaccination. This may temporarily affect the ability to drive or use machines.

4.8 Undesirable effects

Within each frequency grouping, adverse reactions are presented in order of decreasing seriousness. Frequencies are defined as follows:

Very common: ($\geq 1/0$)

Common: ($\geq 1/100$ to $<1/10$)

Uncommon: ($\geq 1/1,000$ to $<1/100$)

Rare: ($\geq 1/10,000$ to $<1/1,000$)

Very rare: ($<1/10,000$)

Not known (cannot be estimated from the available data)

Adverse reactions from clinical trials

Children 2 to 10 years of age

Overall 3,464 subjects aged 2 to 10 years were exposed to Menveo in completed clinical trials. The characterization of the safety profile of Menveo in children 2 to 10 years of age is based on data from four clinical trials in which 3,181 subjects received Menveo.

The most common adverse reactions during the clinical trials generally persisted for one to two days and were not severe. These adverse reactions were:

Metabolism and nutrition disorders:

Very Common: eating disorders

Nervous system disorders:

Very common: sleepiness, headache

Gastrointestinal disorders:

Common: nausea, vomiting, diarrhea

Skin and subcutaneous tissue disorders:

Common: rash

Musculoskeletal and connective tissue disorders:

Common: myalgia, arthralgia

General disorders and administration site conditions:

Very common: irritability, malaise, injection site pain, injection site erythema (≤ 50 mm), injection site induration (≤ 50 mm)

Common: injection site erythema (>50 mm), injection site induration (>50 mm), chills, fever $\geq 38^{\circ}\text{C}$

Uncommon: injection site pruritus

Individuals 11 to 65 years of age

The characterization of the safety profile of Menveo in adolescents and adults is based on data from five randomized controlled clinical trials including 6,401 participants (from 11-65 years of age) who received Menveo. Among Menveo recipients, 58.9%, 16.4%, 21.3% and 3.4% were in the 11-18 year, 19-34 year, 35-55 year and 56-65 year age groups, respectively. The two primary safety studies were randomized, active-controlled trials that enrolled participants aged 11 to 55 years (N=2663) and 19 to 55 years (N=1606), respectively.

The incidence and severity of any, local, systemic, and other reactions were generally similar in the Menveo groups across all studies and within the adolescent and adult age groups. The reactogenicity profile and rates of adverse events among subjects aged 56-65 years who received Menveo (N=216), were similar to that observed in Menveo recipients subjects aged 11-55.

The most common local and systemic adverse reactions observed in clinical trials were pain at the injection site and headache.

The list provided below presents adverse reactions reported in three pivotal and two supportive clinical trials per system organ class. The most common side effects reported during clinical trials usually lasted only one to two days and were not usually severe.

Nervous system disorders:

Very common: headache

Uncommon: dizziness

Gastrointestinal disorders:

Very common: nausea

Skin and subcutaneous tissue disorders:

Common: rash

Musculoskeletal and connective tissue disorders

Very common: myalgia

Common: arthralgia

General disorders and administration site conditions:

Very common: injection site pain, injection site erythema (≤ 50 mm), injection site induration (≤ 50 mm), malaise

Common: injection site erythema (> 50 mm), injection site induration (> 50 mm), fever $\geq 38^{\circ}\text{C}$, chills

Uncommon: injection site pruritus

In the adolescent age group, the safety and tolerability of the vaccine was favourable relative to Tdap and did not substantially change with concomitant or sequential administration of other vaccines.

Post-marketing experience (*all age groups*)

Blood and lymphatic system disorders

Local lymphadenopathy

Immune system disorders

Not known: hypersensitivity including anaphylaxis

Nervous system disorders

Not known: tonic convulsion, febrile convulsion, syncope

Ear and labyrinth disorders

Not known: vertigo

General disorders and administration site conditions

Not known: injection site cellulitis, injection site swelling, including extensive swelling of the injected limb

4.9 Overdose

No case of overdose has been reported.

5. PHARMACOLOGICAL PROPERTIES

5.1 Pharmacodynamic properties

Pharmacotherapeutic group: Meningococcal vaccines, ATC code: J07AH08.

Immunogenicity

The efficacy of Menveo has been inferred by measuring the production of serogroup- specific anti-capsular antibodies with bactericidal activity. Serum bactericidal activity (SBA) was measured using human serum as the source of exogenous complement (hSBA). The hSBA was the original correlate of protection against meningococcal disease.

Immunogenicity was evaluated in randomized, multicenter, active controlled clinical trials that enrolled children (2-10 years of age), adolescents (11-18 years of age) adults (19-55 years of age) and older adults (56-65 years of age).

Immunogenicity in children 2 to 10 years of age)

In the pivotal study V59P20 immunogenicity of Menveo was compared to ACWY-D; 1,170 children were vaccinated with Menveo and 1,161 received the comparator vaccine in the per protocol populations. In two supportive studies V59P8 and V59P10 immunogenicity of Menveo was compared to ACWY-PS.

In the pivotal, randomized, observer-blind study V59P20, in which participants were stratified by age (2 through 5 years and 6 through 10 years), the immunogenicity of a single dose of Menveo one month post vaccination was compared with the single dose of ACWY-D. Immunogenicity results one month after Menveo vaccination among subjects aged 2-5 years and

6-10 years are summarized below in Table 1.

Table 1: Serum bactericidal antibody responses following Menveo one month after vaccination among subjects aged 2-5 years and 6-10 years

Serogroup	2-5 years		6-10 years	
	hSBA $\geq 1:8$ (95% CI)	hSBA GMTs (95% CI)	hSBA $\geq 1:8$ (95% CI)	hSBA GMTs (95% CI)
A	N=606	N=606	N=551	N=551
	72% (68, 75)	26 (22, 30)	77% (74, 81)	35 (29, 42)
C	N=607	N=607	N=554	N=554
	68% (64, 72)	18 (15, 20)	77% (73, 80)	36 (29, 45)
W-135	N=594	N=594	N=542	N=542
	90% (87, 92)	43 (38, 50)	91% (88, 93)	61 (52, 72)
Y	N=593	N=593	N=545	N=545
	76% (72, 79)	24 (20, 28)	79% (76, 83)	34 (28, 41)

In another randomized, observer-blind study (V59P8) US children were immunized with a single dose of either Menveo (N=284) or ACWY-PS (N=285). In the children 2-10 years of age, as well as in each age strata (2-5 and 6-10 years), immune response as measured by percentage of subjects with seroresponse, hSBA $\geq 1:8$ and GMTs were not only non-inferior to comparator vaccine ACWY-PS, but all were statistically higher than the comparator for all serogroups and all immune measurements at 1 month post vaccination. At 1 year post vaccination, Menveo continued to be statistically higher than ACWY-PS for group A, W-135 and Y, as measured by percentage of subjects with hSBA $\geq 1:8$ and GMTs. Menveo was non-inferior on these endpoints for serogroup C (Table 2). The clinical relevance of higher post-vaccination immune responses is not known.

Table 2: Immunogenicity of one dose of Menveo or ACWY-PS in subjects 2 through 10 years of age, measured at one month and twelve months post-vaccination

Serogroup	1 month post-vaccination				12 months post-vaccination			
	hSBA $\geq$ 1:8 (95% CI)		hSBA GMTs (95% CI)		hSBA $\geq$ 1:8 (95% CI)		hSBA GMTs (95% CI)	
	Menveo	ACWY-PS	Menveo	ACWY-PS	Menveo	ACWY-PS	Menveo	ACWY-PS
A	N=280	N=281	N=280	N=281	N=253	N=238	N=253	N=238
	79% (74, 84)	37% (31, 43)	36 (30, 44)	6.31 (5.21, 7.64)	23% (18, 29)	13% (9, 18)	3.88 (3.39, 4.44)	3 (2.61, 3.44)
C	N=281	N=283	N=281	N=283	N=252	N=240	N=252	N=240
	73% (68, 78)	54% (48, 60)	26 (21, 34)	15 (12, 20)	53% (47, 59)	44% (38, 51)	11 (8.64, 13)	9.02 (7.23, 11)
W-135	N=279	N=282	N=279	N=282	N=249	N=237	N=249	N=237
	92% (88, 95)	66% (60, 71)	60 (50, 71)	14 (12, 17)	90% (86, 94)	45% (38, 51)	42 (35, 50)	7.57 (6.33, 9.07)
Y	N=280	N=282	N=280	N=282	N=250	N=239	N=250	N=239
	88% (83, 91)	53% (47, 59)	54 (44, 66)	11 (9.29, 14)	77% (71, 82)	32% (26, 38)	27 (22, 33)	5.29 (4.34, 6.45)

In a randomized, observer-blind study (V59P10) conducted in Argentina, children were immunized with a single dose of either Menveo (N=949) or ACWY-PS (N=551). Immunogenicity was assessed in a subset of 150 subjects in each vaccine group. The immune response observed in the children 2-10 years of age was very similar to those observed in the V59P8 study shown above: immune response to Menveo at 1 month post vaccination, as measured by percentage of subjects with seroresponse, hSBA $\geq$ 1:8 and GMTs, was non-inferior to ACWY-PS.

A randomised, observer-blind study was conducted in children 12 to 59 months of age in Finland and Poland (V59P7). A total of 199 subjects 2-5 years of age were in the Menveo per protocol immunogenicity population and 81 subjects 3-5 years of age were in the ACWY-PS group.

At 1 month post-first vaccination, the percentages of subjects with hSBA $\geq$ 1:8 were consistently higher in the Menveo group for all four serogroups (63% vs 39%, 46% vs 39%, 78% vs 59%, and 65% vs 57% for Menveo as compared to ACWY-PS for serogroups A, C, W-135, and Y, respectively).

In a randomized, observer-blind study (V59_57) conducted in US, immunogenicity of a 2-dose series and a single dose of Menveo was compared in children 2 through 5 and 6 through 10 years of age (N=715).

At baseline, the percentage of subjects with hSBA $\geq$ 1:8 across the two age strata was 1%-5% for serogroup A, 13%-28% for serogroup C, 42%-64% for serogroup W-135, and 6%-19% for serogroup Y. At 1 month post last vaccination, the percentages of subjects with hSBA $\geq$ 1:8 in the 2-dose group and in the single dose group across the two age strata were: 90%-95% vs 76%-80% for serogroup A, 98%-99% vs 76%-87% for serogroup C, 99% vs 93%-96% for serogroup W-135, and 96% vs 65%-69% for serogroup Y. GMTs were higher in the 2-dose group than the

single dose group at 1 month after vaccination in both age strata; however, this difference was less pronounced in the older age stratum.

At 1 year post last vaccination, the percentages of subjects with hSBA $\geq 1:8$ after the 2-dose series and the single dose were both lower than at 1 month post-vaccination (30% after the 2-dose series, 11%-20% after the single dose for serogroup A; 61%-81% and 41%-55% for serogroup C; 92%-94% and 90%-91% for serogroup W-135; 67%-75% and 57%-65% for serogroup Y). The differences between hSBA GMTs in the 2-dose and the single dose groups at 1 year after vaccination were lower than those seen at 1 month post-vaccination.

The clinical benefit of a 2-dose vaccination series in children 2 through 10 years of age is not known.

Persistence of immune response and booster response in children 2 to 10 years of age

Antibody persistence at 5 years after primary vaccination was assessed in study V59P20E1, this was an extension of study V59P20. There was antibody persistence observed against serogroups C, W-135 and Y, with the percentages of subjects with hSBA $\geq 1:8$ being 32% and 56% against serogroup C in subjects 2-5 and 6-10 years of age, respectively, 74% and 80% against serogroup W-135, and 48% and 53% against serogroup Y. GMTs were respectively 6.5 and 12 for serogroup C, 19 and 26 for serogroup W-135, and 8.13 and 10 for serogroup Y. For serogroup A, 14% and 22% of subjects 2-5 and 6-10 years of age, respectively, had hSBA $\geq 1:8$ (GMTs 2.95 and 3.73).

The children also received a booster dose of Menveo, 5 years after a single dose primary vaccination. All subjects in both age groups had hSBA $\geq 1:8$ across all serogroups, with antibody titers several fold higher than seen after the primary vaccination (Table 3).

Table3: Persistence of immune responses 5 years after primary vaccination with Menveo, and immune responses 1 month after a booster dose among subjects aged 2-5 years and 6-10 years at the time of primary vaccination

Serogroup	2-5 years				6-10 years			
	5 year persistence		1 month after booster		5 year persistence		1 month after booster	
	hSBA $\geq 1:8$ (95% CI)	hSBA GMTs (95% CI)	hSBA $\geq 1:8$ (95% CI)	hSBA GMTs (95% CI)	hSBA $\geq 1:8$ (95% CI)	hSBA GMTs (95% CI)	hSBA $\geq 1:8$ (95% CI)	hSBA GMTs (95% CI)
A	N=96	N=96	N=95	N=95	N=64	N=64	N=60	N=60
	14% (7, 22)	2.95 (2.42, 3.61)	100% (96, 100)	361 (299, 436)	22% (13, 34)	3.73 (2.74, 5.06)	100% (94, 100)	350 (265, 463)
C	N=96	N=96	N=94	N=94	N=64	N=64	N=60	N=60
	32% (23, 43)	6.5 (4.75, 8.9)	100% (96, 100)	498 (406, 610)	56% (43, 69)	12 (7.72, 19)	100% (94, 100)	712 (490, 1036)
W-135	N=96	N=96	N=95	N=95	N=64	N=64	N=60	N=60
	74% (64, 82)	19 (14, 25)	100% (96, 100)	1534 (1255, 1873)	80% (68, 89)	26 (18, 38)	100% (94, 100)	1556 (1083, 2237)
Y	N=96	N=96	N=94	N=94	N=64	N=64	N=59	N=59
	48% (38, 58)	8.13 (6.11, 11)	100% (96, 100)	1693 (1360, 2107)	53% (40, 66)	10 (6.51, 16)	100% (94, 100)	1442 (1050, 1979)

Immunogenicity in individuals 11 years of age and above

In the pivotal study (V59P13), participants received either a dose of Menveo (N = 2649) or comparator vaccine (ACWY-D) (N = 875). Sera were obtained both before vaccination and 1 month after vaccination.

In another study (V59P6) conducted in 524 adolescents, the immunogenicity of Menveo was compared to that of ACWY-PS.

Immunogenicity in adolescents

In the 11-18 year old population of the pivotal study, V59P13, the immunogenicity of a single dose of Menveo one month post vaccination is compared with the quadrivalent, ACWY-Diphtheria toxoid protein conjugate vaccine (ACWY-D). Immunogenicity results at one month after Menveo are summarized below in Table 4.

Table 4: Serum bactericidal antibody responses following Menveo one month after vaccination among subjects aged 11-18 years

<i>Serogroup</i>	<i>N</i>	<i>GMT (95% CI)</i>	<i>hSBA ≥ 1:8 (95% CI)</i>
<i>A</i>	1075	29 (24, 35)	75% (73, 78)
<i>C</i>	1396	50 (39, 65)	85% (83, 87)
<i>W-135</i>	1024	87 (74, 102)	96% (95, 97)
<i>Y</i>	1036	51 (42, 61)	88% (85, 90)

In the subset of subjects aged 11-18 years who were seronegative at baseline (hSBA < 1:4), the proportion of subjects who achieved a hSBA ≥ 1:8 after a dose of Menveo were as follows: serogroup A 75% (780/1039); serogroup C 80% (735/923); serogroup W135 94% (570/609); serogroup Y 81% (510/630).

In the non-inferiority study, V59P6, immunogenicity was assessed among adolescents aged 11-17 years who had been randomised to receive either Menveo or ACWY-PS. Menveo was shown to be non-inferior to ACWY-PS vaccine for all four serogroups (A, C, W-135 and Y) based on seroresponse, proportions achieving hSBA ≥ 1:8, and GMTs.

Table 5: Immunogenicity of one dose of Menveo or ACWY-PS in adolescents, measured at one month post vaccination

Serogroup	hSBA ≥ 1:8 (95% CI)		hSBA GMTs (95% CI)	
	Menveo	ACWY-PS	Menveo	ACWY-PS
A	N=140	N=149	N=140	N=149
	81% (74, 87)	41% (33, 49)	33 (25, 44)	7.31 (5.64, 9.47)
C	N=140	N=147	N=140	N=147
	84% (77, 90)	61% (53, 69)	59 (39, 89)	28 (19, 41)
W-135	N=138	N=141	N=138	N=141
	91% (84, 95)	84% (77, 89)	48 (37, 62)	28 (22, 36)
Y	N=139	N=147	N=139	N=147
	95% (90, 98)	82% (75, 88)	92 (68, 124)	35 (27, 47)

At one year post vaccination in these same subjects, compared with ACWY-PS, a higher proportion of subjects vaccinated with Menveo had hSBA ≥ 1:8 for serogroups C, W-135, and Y, with comparable levels for serogroup A. Similar findings were observed in the comparison of hSBA GMTs.

Persistence of immune response and booster response in adolescents

In study V59P13E1, the persistence of immune responses against serogroups A, C, W-135 and Y was assessed at 21 months, 3 years and 5 years post primary vaccination among subjects aged 11-18 years at the time of vaccination. The percentages of subjects with hSBA $\geq 1:8$ remained constant against serogroups C, W-135, and Y from 21 months to 5 years postvaccination in the Menveo group and decreased slightly over time against serogroup A (Table 6). At 5 years after primary vaccination, there were significantly higher percentages of subjects with hSBA $\geq 1:8$ in the Menveo group than in the vaccine naive control subjects against all the four serogroups.

Table 6: Persistence of immune responses approximately 21 months, 3 years and 5 years after vaccination with Menveo (subjects were aged 11-18 years at the time of vaccination)

Serogroup	Timepoint	Percentages of subjects with hSBA $\geq 1:8$	hSBA GMTs
		Menveo	Menveo
A		N=100	N=100
	21 months	45 (35, 55)	6.57 (4.77-9.05)
	3 years	38 (28, 48)	5.63 (3.97-7.99)
	5 years	35 (26, 45)	4.43 (3.13-6.26)
C		N=100	N=100
	21 months	61 (51, 71)	11 (8.12-15)
	3 years	68 (58, 77)	16 (11-25)
	5 years	64 (54, 73)	14 (8.83-24)
W-135		N=99	N=99
	21 months	86 (77, 92)	18 (14-25)
	3 years	85 (76, 91)	31 (21-46)
	5 years	85 (76, 91)	32 (21-47)
Y		N=100	N=100
	21 months	71 (61, 80)	14 (10-19)
	3 years	69 (59, 78)	14 (9.68-20)
	5 years	67 (57, 76)	13 (8.8-20)

A booster dose of Menveo was administered 3 years after primary vaccination with Menveo or ACWY-D. Both groups showed a robust response to the booster dose of Menveo at one month after vaccination

(100% of subjects had hSBA $\geq$ 1:8 across serogroups) and this response largely persisted through 2 years after the booster dose for serogroups C, W-135 and Y (with 87% to 100% of subjects with hSBA $\geq$ 1:8 across serogroups). A small decline was observed in percentages of subjects with hSBA $\geq$ 1:8 against serogroup A, although percentages were still high (77% to 79%). GMTs declined over time as expected but remained between 2- and 8-fold higher than prebooster values (Table 8).

In study V59P6E1, at one year post vaccination, the percentage of Menveo recipients with hSBA $\geq$ 1:8 remained significantly higher compared with ACWY-PS recipients for serogroups C, W-135 and Y, and similar between the two study groups for serogroup A. hSBA GMTs for serogroups W-135 and Y were higher among Menveo recipients. In 5 years post vaccination, the percentage of Menveo recipients with hSBA $\geq$ 1:8 remained significantly higher compared with ACWY-PS recipients for serogroups C and Y. Higher hSBA GMTs were observed for serogroups W-135 and Y (Table 7).

Table 7: Persistence of immune responses approximately 12 months and 5 years after vaccination with Menveo and ACWY-PS (subjects were aged 11-18 years at the time of vaccination)

Serogroup	Timepoint	Percentages of subjects with hSBA $\geq$ 1:8			hSBA GMTs		
		Menveo	ACWY-PS	P Value Menveo vs ACWY-PS	Menveo	ACWY-PS	P Value Menveo vs ACWY-PS
A		N=50	N=50		N=50	N=50	
	12 months	41% (27, 56)	43% (28, 59)	0.73	5.19 (3.34, 8.09)	6.19 (3.96, 9.66)	0.54
	5 years	30% (18, 45)	44% (30, 59)	0.15	5.38 (3.29, 8.78)	7.75 (4.83, 12)	0.24
C		N=50	N=50		N=50	N=50	
	12 months	82% (68, 91)	52% (37, 68)	<0.001	29 (15, 57)	17 (8.55, 33)	0.22
	5 years	76% (62, 87)	62% (47, 75)	0.042	21 (12, 37)	20 (12, 35)	0.92
W-135		N=50	N=50		N=50	N=50	
	12 months	92% (80, 98)	52% (37, 68)	<0.001	41 (26, 64)	10 (6.41, 16)	<0.001
	5 years	72% (58, 84)	56% (41, 70)	0.093	30 (18, 52)	13 (7.65, 22)	0.012
Y		N=50	N=50		N=50	N=50	
	12 months	78% (63, 88)	50% (35, 65)	0.001	34 (20, 57)	9.28 (5.5, 16)	<0.001
	5 years	76% (62, 87)	50% (36, 64)	0.002	30 (18, 49)	8.25 (5.03, 14)	<0.001

A booster dose of Menveo was administered 5 years after primary vaccination with Menveo or ACWY-PS. At 7 days after the booster dose, 98%-100% of subjects who previously received Menveo and 73%-84% of subjects who previously received ACWY-PS achieved hSBA $\geq$ 1:8 against serogroups A, C, W-135 and Y. At one month post vaccination, the percentages of subjects with hSBA $\geq$ 1:8 were 98%-100% and 84%-96%, respectively.

A significant increase in the hSBA GMTs against all four serogroups was also observed at 7 and 28 days after the booster dose (Table 8).

Table 8: Response to Booster: bactericidal antibody responses to Menveo booster administered at 3 or 5 years after the primary vaccination with Menveo or ACWY-PS in subjects aged 11-17 years

Serogroup	Time point	Percentages of subjects with hSBA $\geq$ 1:8			hSBA GMTs		
		V59P13E1 (3 years post vaccination)	V59P6E1 (5 years post vaccination)		V59P13E1 (3 years post vaccination)	V59P6E1 (5 years post vaccination)	
		Menveo	Menveo	ACWY-PS	Menveo	Menveo	ACWY-PS
A		N=42	N=49	N=49	N=42	N=49	N=49
	Pre-booster	21% (10, 37)	29% (17, 43)	43% (29, 58)	2.69 (1.68, 4.31)	5.16 (3.46, 7.7)	7.31 (4.94, 11)
	7 days	-	100% (93, 100)	73% (59, 85)	-	1059 (585, 1917)	45 (25, 80)
	28 days	100% (92, 100)	98% (89, 100)	94% (83, 99)	326 (215, 494)	819 (514, 1305)	147 (94, 232)
	2 years	79% (63, 90)	-	-	22 (12, 41)	-	-
C		N=42	N=49	N=49	N=42	N=49	N=49
	Pre-booster	55% (39, 70)	78% (63, 88)	61% (46, 75)	16 (8.66, 31)	20 (13, 33)	19 (12, 31)
	7 days	-	100% (93, 100)	78% (63, 88)	-	1603 (893, 2877)	36 (20, 64)
	28 days	100% (92, 100)	100% (93, 100)	84% (70, 93)	597 (352, 1014)	1217 (717, 2066)	51 (30, 86)
	2 years	95% (84-99)	-	-	124 (62-250)	-	-
W-135		N=41	N=49	N=49	N=41	N=49	N=49
	Pre-booster	88% (74, 96)	73% (59, 85)	55% (40, 69)	37 (21, 65)	29 (17, 49)	12 (7.02, 19)
	7 days	-	100% (93, 100)	84% (70, 93)	-	1685 (1042, 2725)	34 (21, 54)
	28 days	100% (91, 100)	100% (93, 100)	92% (80, 98)	673 (398, 1137)	1644 (1090, 2481)	47 (32, 71)
	2 years	100% (91, 100)	-	-	93 (58, 148)	-	-
Y		N=42	N=49	N=49	N=42	N=49	N=49
	Pre-booster	74% (58, 86)	78% (63, 88)	51% (36, 66)	14 (8.15, 26)	28 (18, 45)	7.8 (4.91, 12)
	7 days	-	98% (89, 100)	76% (61, 87)	-	2561 (1526, 4298)	21 (13, 35)
	28 days	100% (92, 100)	100% (93, 100)	96% (86, 100)	532 (300, 942)	2092 (1340, 3268)	63 (41, 98)
	2 years	95% (84, 99)	-	-	55 (30, 101)	-	-

Immunogenicity in adults

In the pivotal immunogenicity trial, V59P13, immune responses to Menveo were assessed among adults aged 19 to 55 years. Results are presented in Table 9. In the subset of subjects aged 19-55 years who were seronegative at baseline, the proportion of subjects who achieved a hSBA $\geq 1:8$ after a dose of Menveo were as follows: serogroup A 67% (582/875); serogroup C 71% (401/563); serogroup W-135 82% (131/160); serogroup Y 66% (173/263).

Table 9: Serum bactericidal antibody responses following Menveo one month after vaccination among subjects aged 19-55 years

Serogroup	N	GMT (95% CI)	hSBA $\geq 1:8$ (95% CI)
A	963	31 (27, 36)	69% (66, 72)
C	902	50 (43, 59)	80% (77, 83)
W-135	484	111 (93, 132)	94% (91, 96)
Y	503	44 (37, 52)	79% (76, 83)

The onset of immune response after the primary vaccination with Menveo in healthy subjects 18 through 22 years of age was evaluated in study V59P6E1. At 7 days post vaccination, 64% of subjects achieved hSBA $\geq 1:8$ against serogroup A and 88% through 90% of subjects had bactericidal antibodies against serogroups C, W-135 and Y. At one month post vaccination, 92% through 98% of subjects had hSBA $\geq 1:8$ against serogroups A, C, W-135 and Y. A robust immune response as measured by hSBA GMTs against all serogroups was also observed at 7 days (GMTs 34 through 70) and 28 days (GMTs 79 through 127) after a single dose vaccination.

Immunogenicity in older adults

The comparative immunogenicity of Menveo vs. ACWY-PS was evaluated in subjects aged 56-65 years, in study V59P17. The proportion of subjects with hSBA $\geq 1:8$ was non-inferior to ACWY-PS for all four serogroups and statistically superior for serogroups A and Y (Table 10).

Table 10: Immunogenicity of one dose of Menveo or ACWY-PS in adults aged 56-65 years, measured at one month post vaccination.

Serogroup	Menveo hSBA $\geq$ 1:8 (95% CI)	ACWY-PS hSBA $\geq$ 1:8 (95% CI)
A	N=83	N=41
	87% (78, 93)	63% (47, 78)
C	N=84	N=41
	90% (82, 96)	83% (68, 93)
W-135	N=82	N=39
	94% (86, 98)	95% (83, 99)
Y	N=84	N=41
	88% (79, 94)	68% (52, 82)

Available data in children 2 to 23 months of age

The immunogenicity of Menveo in children 2 to 23 months of age was evaluated in several studies. Although a high percentage of subjects achieved hSBA titres above 1:8 following 4-dose series of Menveo, with lower percentages in studies of 2-dose series and of a single dose, Menveo was compared to another meningococcal vaccine in only one pivotal study, where it failed to show a response at least equivalent to a monovalent conjugated serotype C vaccine (after a single dose at the age of 12 months). Currently available data are not sufficient to establish the efficacy of Menveo in children under 2 years of age. See section 4.2 for information on paediatric use.

5.2 Pharmacokinetic properties

Not applicable.

5.3 Preclinical safety data

Non-clinical data reveal no special hazard for humans based on conventional repeated-dose and reproductive and developmental toxicity studies.

In laboratory animals, no adverse reactions were seen in vaccinated maternal rabbits or in their offspring through postnatal day 29.

No effects on fertility were observed in female rabbits receiving Menveo pre-mating and during pregnancy.

6. PHARMACEUTICAL PARTICULARS

6.1 List of excipients

Powder

Sucrose

Potassium dihydrogen phosphate

Solution

Sodium dihydrogen phosphate monohydrate

Disodium phosphate dihydrate

Sodium chloride

Water for injections

6.2 Incompatibilities

This medicinal product must not be mixed with other medicinal products except those mentioned in section 6.6.

6.3 Shelf life

4 years.

After reconstitution, the product should be used immediately. However, chemical and physical stability after reconstitution was demonstrated for 8 hours below 25°C.

6.4 Special precautions for storage

Store in a refrigerator (2°C - 8°C). Do not freeze.

Keep the vial(s) and/or syringe in the outer carton in order to protect from light.

For storage conditions of the reconstituted product, see section 6.3.

6.5 Nature and contents of container

Vial-Syringe presentation

Powder in vial (type I glass) with a stopper (butyl rubber) and solution in pre-filled syringe (type I glass) with a tip cap (type I elastomeric closure with natural rubber latex or type II elastomeric closure that has no detectable natural rubber latex).

Each pack contains a single dose of one vial and one pre-filled syringe.

Vial-Vial presentation

Powder in vial (type I glass) with a stopper (butyl rubber) and solution in vial (type I glass) with a stopper (butyl rubber). The contents of the two components (powder vial and solution vial) are to be mixed prior to vaccination providing one dose of 0.5 ml.

Pack size of one dose (2 vials) or five doses (10 vials).

Not all pack sizes may be marketed.

6.6 Special precautions for disposal and other handling

The contents of the two components in the two different containers (MenA powder and MenCWY solution) are to be mixed prior to vaccination providing 1 dose of 0.5 ml.

Vial-Syringe presentation:

Menveo must be prepared for administration by reconstituting powder (in vial) with solution (in pre-filled syringe).

The components of the vaccine should be visually inspected before and after reconstitution.

Remove the tip cap from the syringe and attach a suitable needle for the withdrawal (21G, 1 ½ inch length or a 21G, 40 mm length). Use the whole contents of the syringe (0.6 ml) to reconstitute the powder.

Invert and shake the vial vigorously and then withdraw 0.5 ml of reconstituted product. Please note that it is normal for a small amount of liquid to remain in the vial following withdrawal of the dose.

Following reconstitution, the vaccine is a clear, colourless to light yellow solution, free from visible foreign particles. In the event of any foreign particulate matter and/or variation of physical aspect being observed, discard the vaccine.

Prior to injection, change the needle for one suitable for the administration. Ensure that no air bubbles are present in the syringe before injecting the vaccine.

Vial-Vial presentation:

Menveo must be prepared for administration by reconstituting powder (in vial) with solution (in vial).

The components of the vaccine should be visually inspected before and after reconstitution.

Using a syringe and a suitable needle (21G, 1 ½ inch length or a 21G, 40 mm length) withdraw the entire contents of the vial of solution and inject into the vial of powder to reconstitute the MenA conjugate component.

Invert and shake the vial vigorously and then withdraw 0.5 ml of reconstituted product. Please note that it is normal for a small amount of liquid to remain in the vial following withdrawal of the dose.

Following reconstitution, the vaccine is a clear, colourless to light yellow solution, free from visible foreign particles. In the event of any foreign particulate matter and/or variation of physical aspect being observed, discard the vaccine.

Prior to injection, change the needle for one suitable for the administration. Ensure that no air bubbles are present in the syringe before injecting the vaccine.

For all presentations:

Previously frozen product should not be used. For a description on use, see section SHELF LIFE.

Special Precautions for disposal

Any unused medicinal product or waste material should be disposed of in accordance with local requirements.

7. MANUFACTURER

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